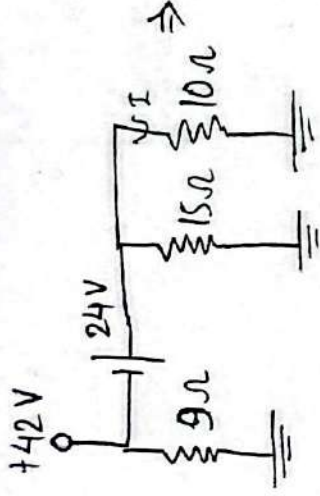
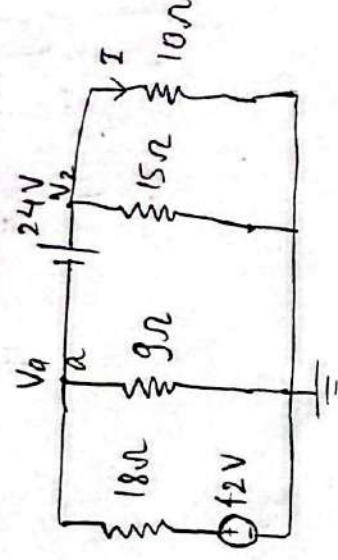




1)



At node a,

$$\frac{V_a - 42}{18} + \frac{V_a}{9} + \frac{V_a}{15} + \frac{V_a}{10} = 0$$

$$\text{or, } 5a - 216 + 10V_a + 6V_a + 4V_a = 0$$

$$\text{or, } 30V_a = 216$$

$$\therefore V_a = 7.2 \text{ V}$$

$$I' = \frac{V_a}{10} = 0.72 \text{ A}$$

For supernode,

$$\frac{V_1}{18} + \frac{V_1}{9} + \frac{V_2}{15} + \frac{V_2}{10} = 0$$

$$5V_1 + 10V_1 + 6V_2 + 9V_2 = 0$$

$$15V_1 + 15V_2 = 0 \quad \text{--- (1)}$$

$$V_2 - V_1 = 24$$

$$V_2 = 24 + V_1 \quad \text{--- (2)}$$

$$15V_1 + 15(24 + V_1) = 0$$

$$15V_1 + 360 + 15V_1 = 0$$

$$30V_1 = -360$$

$$V_1 = -12 \text{ V}$$

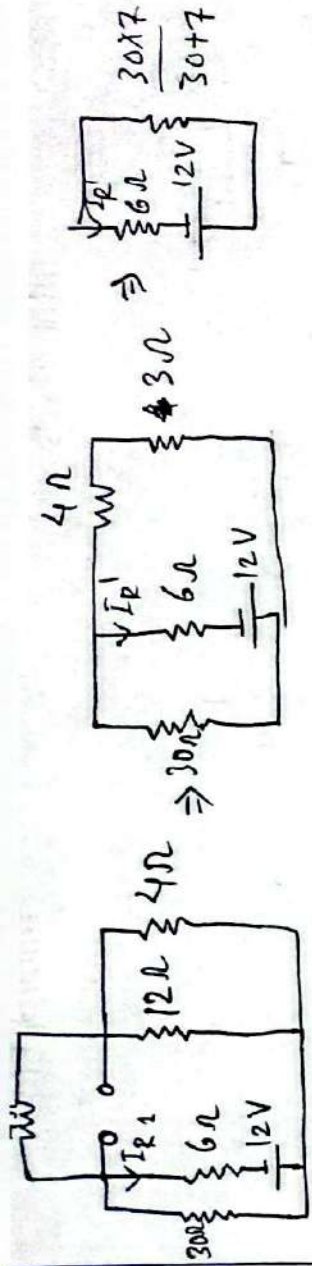
$$V_2 = 24 - 12 = +12 \text{ V}$$

$$I'' = \frac{V_2}{10} = \frac{12}{10} = 1.2 \text{ A}$$

From superposition theorem,

$$I = 0.72 + 1.2 = 2.1 \text{ A}$$

2)



$$\therefore I_{R1}' = \frac{12}{6 + \frac{20}{37}} = 1.0277 \text{ A}$$

considering 6 A source alone,

$$I = \frac{6 \times 4}{8 + 4} = 2 \text{ A}$$

$$I_{R1}'' = \frac{30 \times 2}{30 + 6} = \frac{5}{3} = 1.666 \text{ A}$$

considering 8 V source alone,

$$I = \frac{8}{64/7} = 0.875 \text{ A}$$

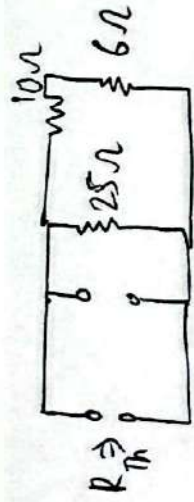
$$I_{R1}' = \frac{12 \times 7/8}{12 + 9} = 0.5 \text{ A}$$

$$I_{R1}''' = \frac{30 \times I_1}{30 + 6} = \frac{30 \times 0.5}{36} = 0.4166$$

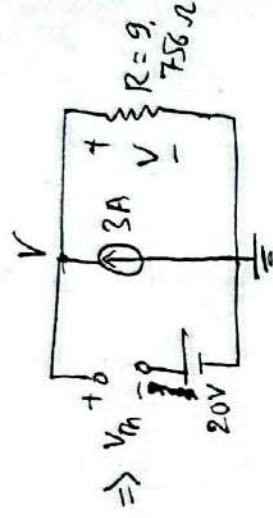
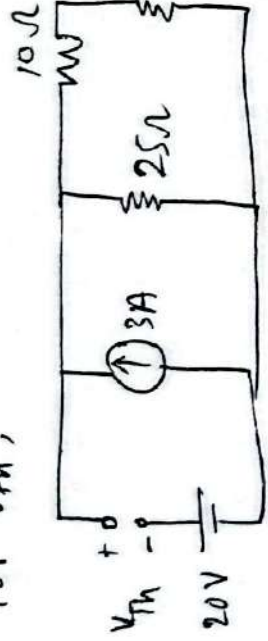
Using superposition theorem,

$$I_{R1} = I_{R1}' + I_{R1}'' + I_{R1}''' = 1.0277 + 1.666 + 0.4166 = 3.11 \text{ A}$$

(3)



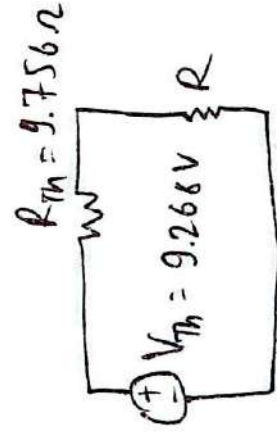
$$R_{th} = \frac{25 \times 6}{25 + 6} = 9.756 \Omega$$

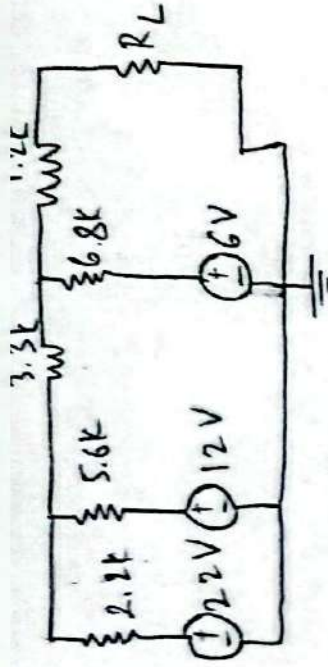
For V_{th} ,

$$\therefore V = 3 \times \frac{25 \times 6}{25 + 6} = 29.268$$

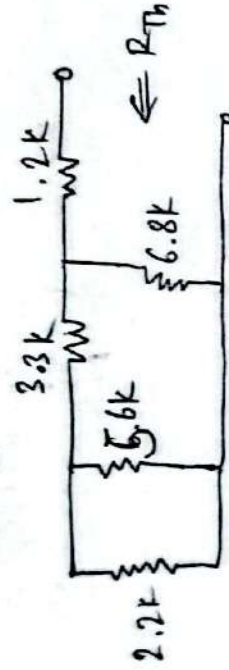
$$V_{th} = V - 20 = 9.268 \text{ V}$$

$\therefore$ The thevenin equivalent circuit is





↓
FOR R_{Th}



$$\therefore R_{Th} = \left(\frac{2.2 \times 5.6}{2.2 + 5.6} + 3.3 \right) 6.8 + 1.2 = 4.041 k\Omega$$

$$\frac{2.2 \times 5.6}{2.2 + 5.6} + 3.3 + 6.8$$

FOR V_{Th}

$$\frac{V_1 - 12}{2.2} + \frac{V_1 + 12}{5.6} + \frac{V_1 - 6}{10.1} = 0$$

$$\text{or, } 56.56(V_1 - 12) + 22.22(V_1 + 12) + 12.32(V_1 - 6) = 0$$

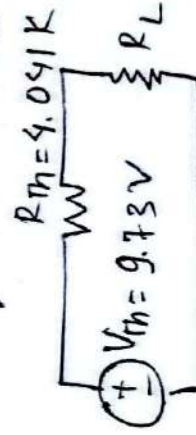
$$\text{or, } 55.56V_1 + 22.22V_1 + 1244.32 + 266.64 + 12.32V_1 - 73.92 = 0$$

$$\text{or, } 91.1V_1 = 1051.6$$

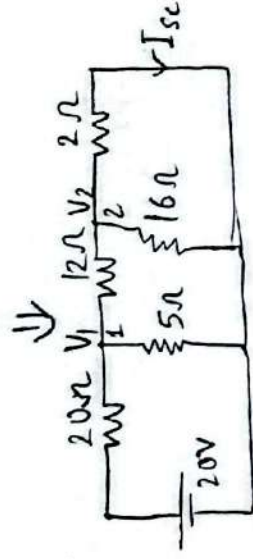
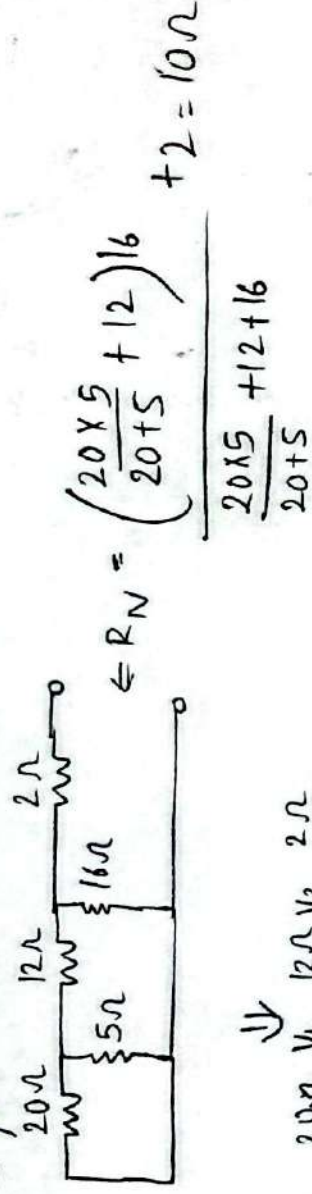
$$\therefore V_1 = 11.543 V$$

$$V_{Th} = 6 + 6.8 \times \frac{11.543}{3.3 + 6.8} = 9.73 V$$

$\therefore$ The thevenin equivalent network is



5(a) To find Norton resistance



At Node 1:

$$\frac{V_1 - 20}{20} + \frac{V_1}{5} + \frac{V_1 - V_2}{12} = 0$$

$$3V_1 - 60 + 12V_1 + 5V_1 - 5V_2 = 0$$

$$20V_1 - 5V_2 = 60$$

$$4V_1 - V_2 = 12 \quad \text{--- (1)}$$

At node 2,

$$\frac{V_2 - V_1}{12} + \frac{V_2}{16} + \frac{V_2}{2} = 0$$

$$4V_2 - 4V_1 + 3V_2 + 24V_2 = 0$$

$$31V_2 - 4V_1 = 0$$

$$V_1 = 31/4 V_2 \quad \text{--- (2)}$$

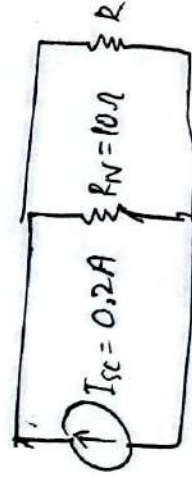
from (1) & (2),

$$4 \times \frac{31}{4} V_2 - V_2 = 12$$

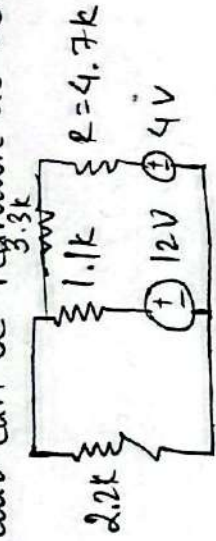
$$V_2 = 0.4V$$

$$I_{sc} = \frac{V_2}{2} = 0.2A$$

Norton equivalent circuit is

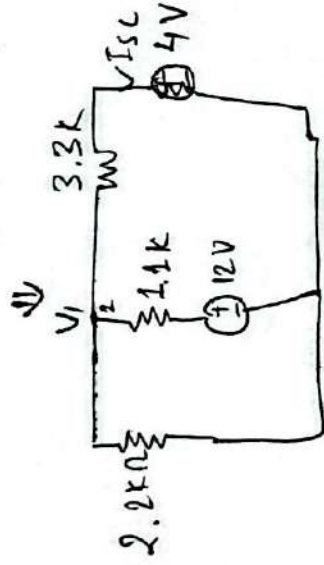


5(b) The circuit can be redrawn as below,



For R_N

$$R_N = \frac{2.2 \times 1.1}{2.2 + 1.1} + 3.3 = 4.033 \text{ k}\Omega$$



At node 1:

$$\frac{V_1}{2.2} + \frac{V_1 - 12}{1.1} + \frac{V_1 + 4}{3.3} = 0$$

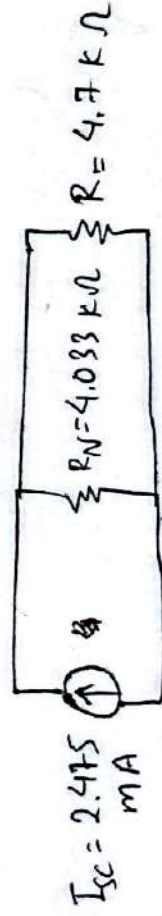
$$\text{or, } 3V_1 + 6V_1 - 72 + 2V_1 + 8 = 0$$

$$\text{or, } 11V_1 = 64$$

$$\therefore V_1 = +5.818 \text{ V}$$

$$I_{SC} = \frac{V_1 + 4}{3.3 \times 10^3} = 2.975 \text{ mA}$$

Norton equivalent circuit is

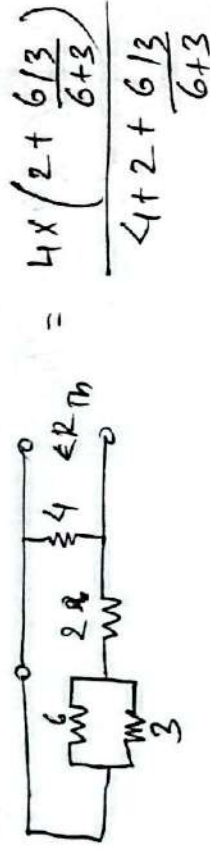


6 (a) For max power transfer

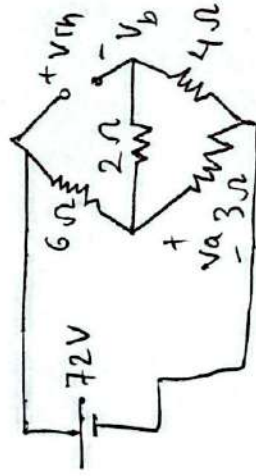
$$R = R_{Th}$$

$$\text{Max power to } R = \frac{V_{Th}^2}{4R_{Th}}$$

For R_{Th}



For V_{Th} ,



$$V = \frac{72 \times 6}{6+2} = 54V$$

$$V_a = \frac{72 \times 2}{8} = 18V$$

$$V_b = \frac{2 \times 18}{6} = 6V$$

$$\therefore V_{Th} = V_a + V_b$$

$$= 54 + 6 = 60V$$

$$\text{Max power} = \frac{V_{Th}^2}{4R_{Th}}$$

$$= \frac{60 \times 60}{4 \times 2}$$

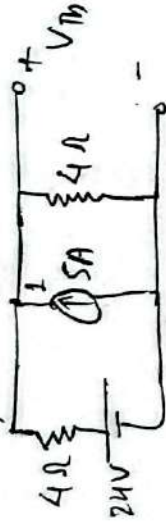
$$= 450W$$

6b)

To find R_{Th} ,

$$4\Omega \parallel 4\Omega \Leftarrow R_{Th} = \frac{4 \times 4}{4 + 4} = 2\Omega$$

To find V_{Th} ,



At node 1,

$$\frac{V_{Th} - 24}{4} + \frac{V_{Th}}{4} - 5 = 0$$

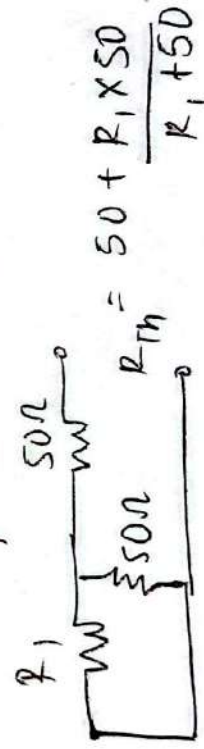
$$V_{Th} - 24 + V_{Th} - 20 = 0$$

$$2V_{Th} = 44 \text{ V}$$

$$\therefore V_{Th} = 22 \text{ V}$$

$$\text{For max power} = \frac{V_{Th}^2}{4R_{Th}} = \frac{22^2}{4 \times 2} = 60.5 \text{ W}$$

(7) For maximum power transfer to R_L the Thevenin resistance of the circuit should be equal to R_L .



For maximum power transfer,

$$R_{Th} = R_L$$

$$50 + R_1 \times 50 = 50$$

$$\therefore R_1 = 0 \Omega$$